

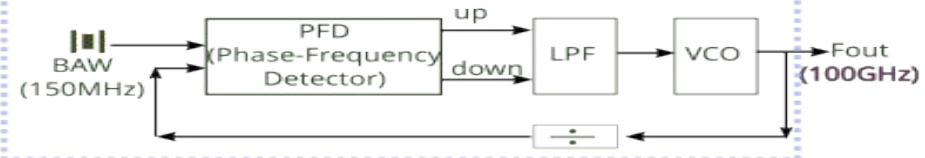
200G PAM4 DCI Architecture

A Deep Dive into 3nm CMOS SerDes Design, High-Speed Clocking,
and Concatenated FEC Simulation

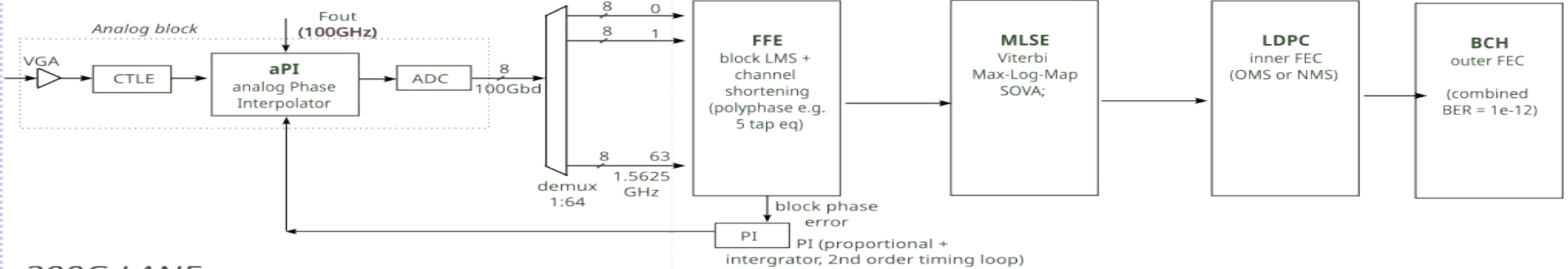
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01. Lane Architecture

Frequency synthesis (simplified, shared by all the lanes)



DSP @ 1.5625GHz

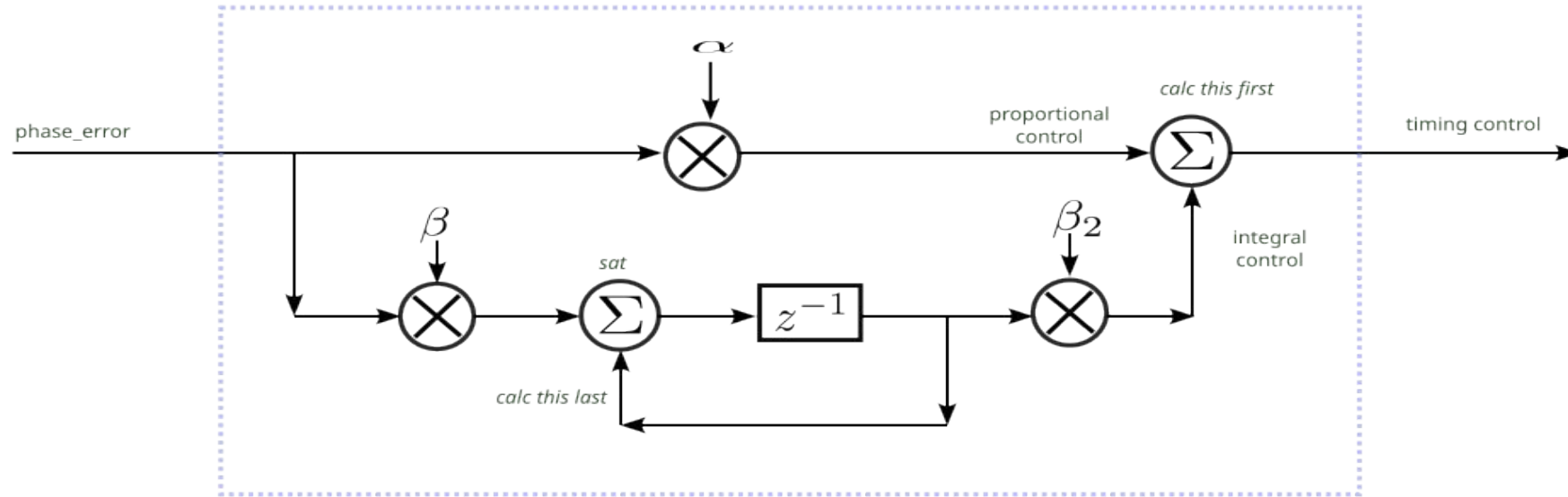


200G LANE

200G LANE

Timing Block

(PI block - proportional integrator)



β_2

Bit-Growth Management: In hardware, β_2 is often just a right-shift ($\gg N$) to make the error very small control before adding it to a wide, high-precision accumulator register. β_2 is then used to truncate or scale that wide accumulator down to the required bit-width for the output control signal.

Unit Conversion: Sometimes the accumulator holds pure "Fractional UI", but the hardware Phase Interpolator requires a specific integer code (e.g., 0 to 63 for one UI). β_2 acts as the hardware mapping gain (e.g., multiplying the fractional UI by 64)

02. FEC Strategy

INNER CODE: LAYERED LDPC

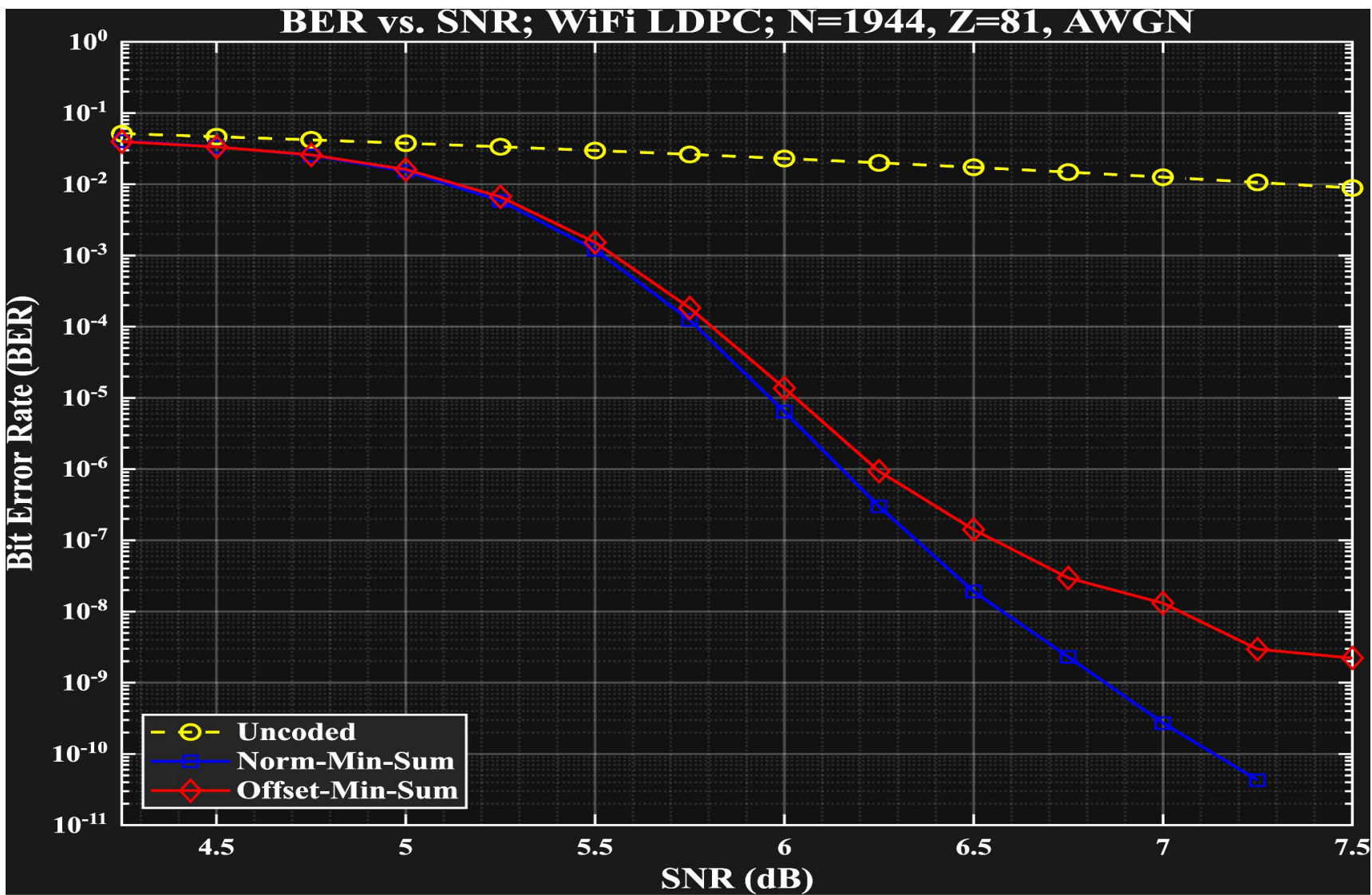
40+
MBPS JAX THROUGHPUT

Soft-Decision Decoding

Inner Layered Belief Propagation (BP) decoder implemented in JAX/XLA. The model leverages **float16** state storage and **float32** compute to optimize RTX 3090 memory bandwidth.

$$Q_{v \rightarrow c} = L_v - R_{\text{old}, c \rightarrow v}$$

LDPC Reference BER vs SNR



OUTER CODE: BCH CLEANUP

- 🔧 **Error Floor Mitigation:** Algebraic BCH decoding is used to clean up residual bit errors from LDPC "trapping sets" and "oscillations."
- ⚡ **Low Latency:** BCH decoders provide deterministic latency, critical for DCI links that cannot afford iterative decoding variance.
- 🛡️ **Trapping Set Escape:** Quantization in JAX (float16) introduces numerical "regularization" that helps the decoder escape sub-optimal local minima.

CONCATENATED PERFORMANCE



Net Coding Gain

The combination of LDPC(1944, 5/6) and BCH(2048, 1944) delivers >11 dB NCG at $1e-12$ BER floors.



JAX Scale-Up

Transitioning from 3090 single-GPU to A100 multi-GPU via `jax.pmap` allows simulation of $1e-15$ BER targets.



MATLAB Verification

Perfect alignment with MATLAB 'approxIir' math verifies the PRNG and AWGN channel noise floor.

03. Timing Recovery

CONCLUSION & Q&A

3djax Project: GPU-Accelerated FEC Simulations.

Architecture: 3nm Monolithic 200G SerDes.

Innovation: JAX-based static graph optimization for high-speed hardware modeling.

